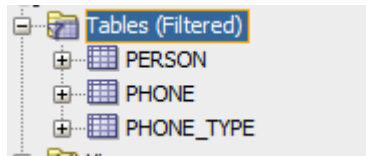


LAB 5

1. Creation of procedure (using OPEN of the cursor) that returns all the phone numbers of a person.

TABLES:



CURSOR

```

Welcome Page x 2_TEST_PROCEDURE_GETPHONES.sql x PHONE x GETPHONES x
Code Grants Dependencies Details Profiles Errors References

create or replace PROCEDURE getPhones(pIdPerson IN NUMBER, pTypeName IN VARCHAR2 DEFAULT NULL)
AS
    CURSOR phoneCursor IS
        SELECT p.first_name, p.first_lastname, ph.phone_number, pt.type_name
        FROM person p
        INNER JOIN phone ph
        ON p.id_person = ph.id_person
        INNER JOIN phone_type pt
        ON ph.id_type = pt.id_type
        WHERE p.id_person = pIdPerson
        AND pt.type_name = NVL(pTypeName, pt.type_name);

    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    OPEN phoneCursor;
    DBMS_OUTPUT.PUT_LINE(LPAD('Name', 20) || LPAD('Phone', 20) || LPAD('Type', 20));
    LOOP
        FETCH phoneCursor INTO vFirstName, vLastName, vnPhoneNumber, vTypeName;
        EXIT WHEN phoneCursor%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD(vTypeName, 20));
    END LOOP;
    CLOSE phoneCursor;
END getPhones;

```

2. Creation of test to test the procedure using cursors.

- PERSON TABLE

Columns	Data	Model	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
ID_PERSON	FIRST_NAME	SECOND_NAME	FIRST_LASTNAME	SECOND_LASTNAME								
1	1 Diego	Josue	Mora	Araya								
2	2 Elena	Maria	Diaz	Mora								

- PHONE TABLE

Columns	Data	Model	Constraints	Grants	Statistics	Triggers	Flashback
Sort.. Filter:							
	ID_PHONE	ID_PERSON	ID_TYPE	PHONE_NUMBER			
1	3	1	1	22512390			
2	4	2	1	23515921			
3	5	1	2	89512342			
4	6	1	3	67576373			
5	7	1	3	60518043			
6	8	2	3	80519099			
7	9	2	3	80519099			
8	10	2	2	81819039			
9	11	2	3	80001993			

- PHONE_TYPE TABLE

Columns	Data	Model	Constraints	Grants	Statistics
Sort.. Filter:					
	ID_TYPE	TYPE_NAME			
1	1	Home			
2	2	Office			
3	3	Personal			

TESTS

SQL Worksheet History

Worksheet Query Builder

```
-- ++++++  
  
-- It is executed once to display the prints in the console  
SET SERVEROUTPUT ON;  
  
--1 _____  
  
DECLARE  
    pIdPerson NUMBER := 2;  
    pTypeName VARCHAR2(20) := 'Home';  
BEGIN  
    getPhones(pIdPerson, pTypeName);  
END;  
  
--2 _____
```

Script Output x

Task completed in 0.149 seconds

Name	Phone	Type
Elena Diaz	23515921	Home

PL/SQL procedure successfully completed.

```
--2  
  
DECLARE  
    pIdPerson NUMBER := 1;  
    pTypeName VARCHAR2(20) := NULL;  
BEGIN  
    getPhones(pIdPerson, pTypeName);  
END;
```

Script Output x

Task completed in 0.169 seconds

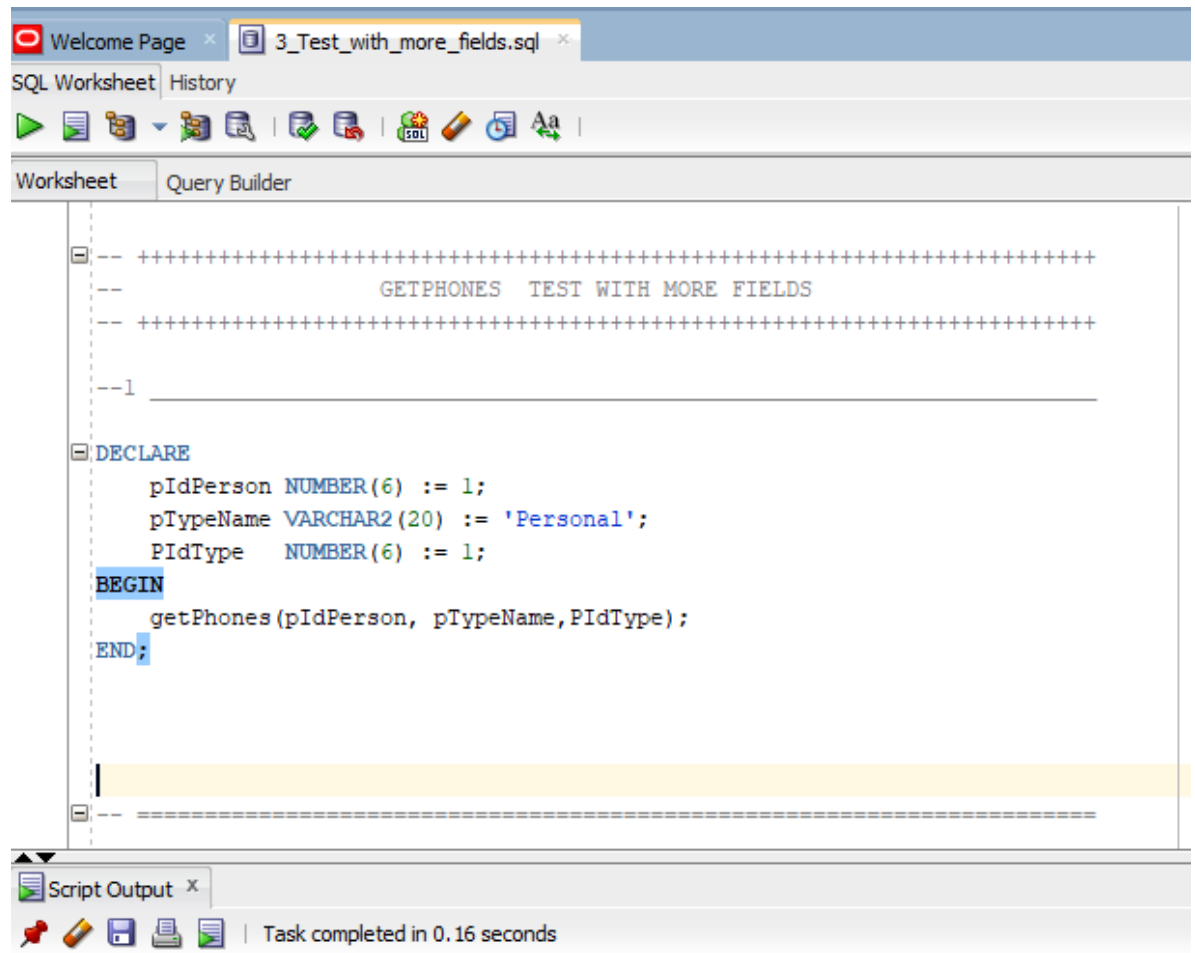
Name	Phone	Type
Diego Mora	22512390	Home
Diego Mora	89512342	Office
Diego Mora	67576373	Personal
Diego Mora	60518043	Personal

PL/SQL procedure successfully completed.

3. What happens when more fields are added in the test fetch?

TEST

Error when the cursor procedure is called with more arguments than the ones established in the GETPHONES parameters(When the cursor test fetch has not been modified):



```

Error starting at line : 8 in command -
DECLARE
    pIdPerson NUMBER(6) := 1;
    pTypeName VARCHAR2(20) := 'Personal';
    PidType    NUMBER(6) := 1;
BEGIN
    getPhones(pIdPerson, pTypeName, PidType);
END;
Error report -
ORA-06550: line 6, column 5:
PLS-00306: wrong number or types of arguments in call to 'GETPHONES'
ORA-06550: line 6, column 5:
PL/SQL: Statement ignored
06550. 00000 - "line %s, column %s:\n%s"
*Cause:    Usually a PL/SQL compilation error.
*Action:

```

Error when more fields are put into the FETCH:

The screenshot shows the SQL Developer interface with a PL/SQL procedure named `getPhones` and its execution errors.

```
create or replace PROCEDURE getPhones(pIdPerson IN NUMBER,pTypeName IN VARCHAR2 DEFAULT NULL)
AS
    CURSOR phoneCursor IS
        SELECT p.first_name, p.first_lastname, ph.phone_number, pt.type_name
        FROM person p
        INNER JOIN phone ph
        ON p.id_person = ph.id_person
        INNER JOIN phone_type pt
        ON ph.id_type = pt.id_type
        WHERE p.id_person = pIdPerson
        AND pt.type_name = NVL(pTypeName, pt.type_name);

    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
    vSecondName person.second_name%TYPE;
BEGIN
    OPEN phoneCursor;
    DBMS_OUTPUT.PUT_LINE(LPAD('Name', 20) || LPAD('Phone', 20) || LPAD('Type', 20));
    LOOP
        FETCH phoneCursor INTO vFirstName, vLastName, vnPhoneNumber, vTypeName, vSecondName;
        EXIT WHEN phoneCursor%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD(vTypeName, 20));
    END LOOP;
    CLOSE phoneCursor;
END getPhones;
```

Project: C:\Users\dauid\AppData\Roaming\SQL Developer\system22.2.1.234.1810\o.sqldeveloper\projects\IdeConnections#ge.jpr

Procedure GE.GETPHONES@ge

- Error(22,9): PL/SQL: SQL Statement ignored
- Error(22,9): PLS-00394: wrong number of values in the INTO list of a FETCH statement

4. What happens when the fetch of the test failed to obtain fields?

TEST

Error when the cursor procedure is called with fewer arguments than the ones established in the GETPHONES parameters(When the cursor test fetch has not been modified):

The screenshot shows an SQL Worksheet window with the following tabs: Welcome Page, Statements - Log, 4_Test_with_fewer_fields.sql, and GETPHONES. The main area displays a PL/SQL script. The script starts with a comment block, followed by a DECLARE statement for pIdPerson, a BEGIN block, a call to getPhones(), and an END statement. The Script Output pane at the bottom shows the execution results, including an error message.

```
-- +-----+
--                                     GETPHONES  TEST WITH FEWER FIELDS
-- +-----+

--1

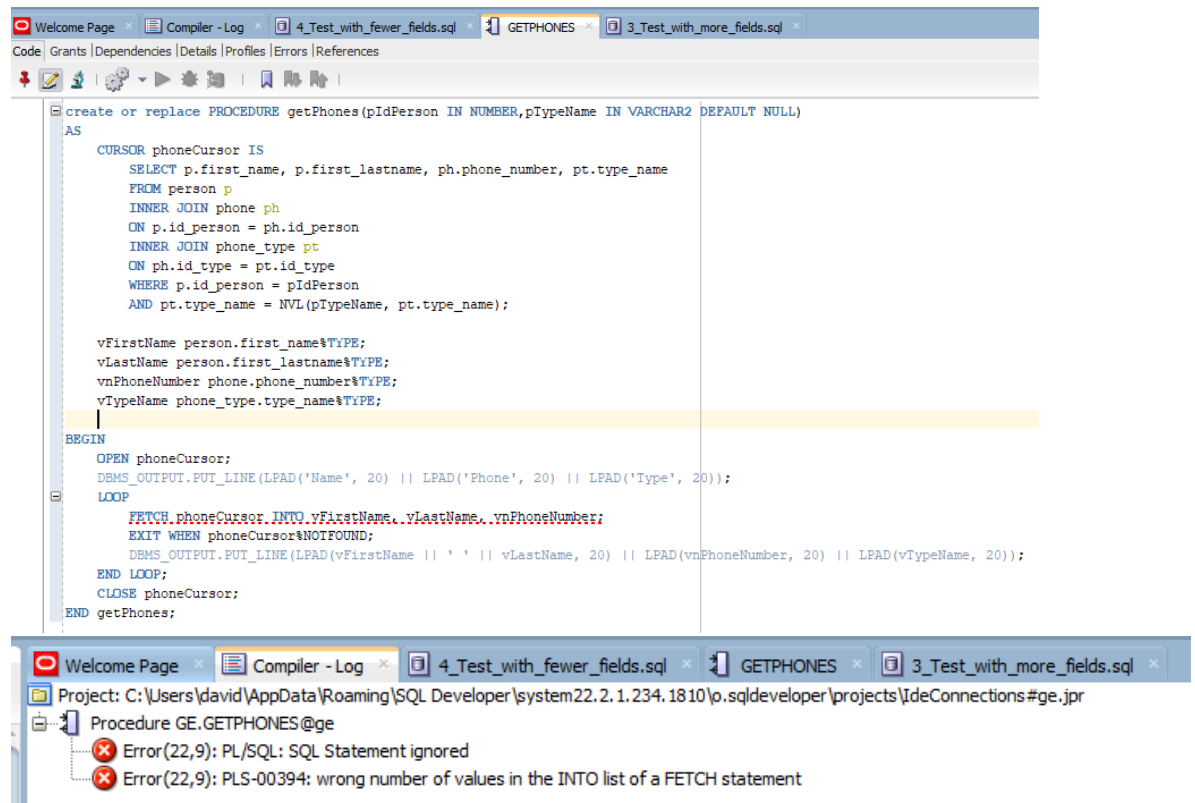
DECLARE
    pIdPerson NUMBER(6) := 1;
BEGIN
    getPhones();
END;
```

Script Output x

Task completed in 0.083 seconds

Error starting at line : 7 in command -
DECLARE
 pIdPerson NUMBER(6) := 1;
BEGIN
 getPhones();
END;
Error report -
ORA-06550: line 4, column 5:
PLS-00306: wrong number or types of arguments in call to 'GETPHONES'
ORA-06550: line 4, column 5:
PL/SQL: Statement ignored
06550. 00000 - "line %s, column %s:\n%s"
*Cause: Usually a PL/SQL compilation error.
*Action:

Error when fewer fields are placed in the FETCH:



5. What happens when in the test fetch the fields have different domains from the what is getting the cursor?

TEST

The fields have different domains from the what is getting the cursor(When the cursor test fetch has not been modified).

SQL Worksheet History

Worksheet Query Builder

```

-- ++++++
--                               GETPHONES TEST WITH DIFFERENT DOMAIN
-- ++++++

DECLARE
  pIdPerson VARCHAR2(20) := '1';
  pTypeName VARCHAR2(20) := 'Oficce';
BEGIN
  getPhones(pIdPerson, ptypename);
END;

```

-- What happens when in the test fetch the fields have different domains
 -- from the what is getting the cursor?
 -- -----

--The procedure doesn't return anything

Script Output x

Task completed in 0.08 seconds

Name	Phone	Type
PL/SQL procedure successfully completed.		

What happens when in the test fetch the fields have different domains from the what is getting the cursor?

The screenshot shows an SQL IDE with the following components:

- Top Bar:** Contains tabs for 'Welcome Page', 'ge', '5_Test_with_fields_of_different_domain_to_the_cursor.sql', '1_CURSOR_GETPHONES.sql', and '2_TE'.
- SQL Worksheet:** Displays the PL/SQL code for a procedure named `getPhones`. The code includes a `DECLARE` section with variables `pIdPerson` and `pTypeName`, a `BEGIN` section with a call to `getPhones`, and an `END;` statement. The code is surrounded by comments indicating it's a test with different domains.
- Script Output:** Shows the execution results. It includes a header with columns 'Name', 'Phone', and 'Type'. Below the header, it states 'PL/SQL procedure successfully completed.' and 'Task completed in 0.057 seconds'.
- Messages - Log:** Shows a log of messages, including 'Compiled' and 'Compiled'.

```

-- +-----+
-- GETPHONES TEST WITH DIFFERENT DOMAIN
-- +-----+

DECLARE
  pIdPerson NUMBER(6) := 1;
  pTypeName VARCHAR2(20) := 'Oficce';
BEGIN
  getPhones(pIdPerson,pTypeName);
END;
```

-- What happens when in the test fetch the fields have different domains
-- from the what is getting the cursor?

--The procedure doesn't return anything

Script Output x

Task completed in 0.057 seconds

Name	Phone	Type
PL/SQL procedure successfully completed.		

Messages - Log x

Welcome Page x ge x GETPHONES x 5_Test_with_fields_of_different_domain_to_the_cursor.sql x

Compiled

Compiled

6. Changing the cursor to use the code with LOOP.

CURSOR

```

create or replace PROCEDURE getPhonesWithLoop(pIdPerson IN NUMBER, pTypeName IN VARCHAR2 DEFAULT NULL)
AS
    CURSOR phoneCursorL
    IS
        SELECT p.first_name, p.first_lastname, ph.phone_number, pt.type_name
        FROM person p
        INNER JOIN phone ph
        ON p.id_person = ph.id_person
        INNER JOIN phone_type pt
        ON ph.id_type = pt.id_type
        WHERE p.id_person = pIdPerson
        AND pt.type_name = NVL(pTypeName, pt.type_name);
BEGIN
    DBMS_OUTPUT.PUT_LINE(LEAD('Name', 20) || LPAD('Phone', 20) || LPAD('Type', 20));
    FOR i IN phoneCursorL LOOP
        DBMS_OUTPUT.PUT_LINE(LEAD(i.first_name || ' ' || i.first_lastname, 20) || LPAD(i.phone_number, 20) || LPAD(i.type_name, 20));
    END LOOP;
END getPhonesWithLoop;

```

TESTS

SQL Worksheet | History

Worksheet | Query Builder

```

--1 _____

DECLARE
    vIdPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := 'Home';

BEGIN
    getPhonesWithLoop(vIdPerson, vcTypeName);
END; -- se agrega esta línea para cerrar el bloque BEGIN.

--2 _____

DECLARE

```

Script Output x

Task completed in 0.111 seconds

Name	Phone	Type
Diego Mora	22512390	Home

PL/SQL procedure successfully completed.

Compiler - Log x Welcome Page x 1_CURSOR_GETPHONES.sql x 6_Modify_Phone_Cursor_with_Loop.sql x

SQL Worksheet History

Worksheet Query Builder

```
--2 _____
DECLARE
    vIdPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := NULL;

BEGIN
    getPhonesWithLoop(vIdPerson,vcTypeName);

END;
```

```
--3 _____
DECLARE
    vIdPerson NUMBER := 1;
```

Script Output x

Task completed in 0.15 seconds

Name	Phone	Type
Diego Mora	22512390	Home
Diego Mora	89512342	Office
Diego Mora	67576373	Personal
Diego Mora	60518043	Personal

PL/SQL procedure successfully completed.

Compiler - Log x Welcome Page x SQL ge x 1_CURSOR_GETPHONES.sql x 6_Modify_Phone_Cursor_with_Loop.sql x

SQL Worksheet History

Worksheet Query Builder

```
--3  
  
DECLARE  
    vIdPerson NUMBER(6) := 1;  
    vcTypeName VARCHAR2(20) := 'Office';  
  
BEGIN  
    getPhonesWithLoop(vIdPerson,vcTypeName);  
END;  
  
--4  
  
DECLARE  
    vIdPerson NUMBER := 1;
```

Script Output x

Task completed in 0.108 seconds

Name	Phone	Type
Diego Mora	89512342	Office

PL/SQL procedure successfully completed.

Compiler - Log Welcome Page ge 1_CURSOR_GETPHONES.sql 6_Modify_Phone_Cursor_with_Loop.sql

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vIdPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := 'Office';

BEGIN
    getPhonesWithLoop(vIdPerson, vcTypeName);
END;

--4

DECLARE
    vIdPerson NUMBER(6) := 2;
    vcTypeName VARCHAR2(20) := 'Personal';

BEGIN
    getPhonesWithLoop(vIdPerson, vcTypeName);
END;

```

Script Output x

Task completed in 0.102 seconds

Name	Phone	Type
Elena Diaz	80519099	Personal
Elena Diaz	80519099	Personal
Elena Diaz	80001993	Personal

PL/SQL procedure successfully completed.

7. Changing the cursor to use the code with SYS_REFCURSOR.

CURSOR

Compiler - Log Welcome Page ge 7_Modify_Phone_Cursor_with_SYS_REFCURSOR.sql 1_CURSOR_GETPHONES.sql GETPHONESWITHSYSREFCURSOR

Code References Profiles Grants Details Errors Dependencies

```

create or replace PROCEDURE getPhonesWithSysRefCursor(pIdPerson IN NUMBER,
    pTypeName IN VARCHAR2 DEFAULT NULL, pRecordSet OUT SYS_REFCURSOR)
AS
    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    OPEN pRecordSet FOR
    SELECT p.first_name, p.first_lastname, ph.phone_number, pt.type_name
    FROM person p
    INNER JOIN phone ph
    ON p.id_person = ph.id_person
    INNER JOIN phone_type pt
    ON ph.id_type = pt.id_type
    WHERE p.id_person = pIdPerson
    AND pt.type_name = NVL(pTypeName, pt.type_name);

END getPhonesWithSysRefCursor;

```

TESTS

Compiler - Log Welcome Page ge 7_Modify_Phone_Cursor_with_SYS_REFCURSOR.sql 1_CURSOR_GETPHONES.sql

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vidPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := 'Home';
    phoneCursorSys SYS_REFCURSOR;
    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    getPhonesWithSysRefCursor(vidPerson, vcTypeName, phoneCursorSys);
    LOOP
        FETCH phoneCursorSys INTO vFirstName, vLastName, vnPhoneNumber, vTypeName;
        EXIT WHEN phoneCursorSys%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD(vTypeName, 20));
    END LOOP;
    CLOSE phoneCursorSys;
END;

```

Script Output x

Task completed in 0.064 seconds

Diego Mora	22512390	Home
------------	----------	------

PL/SQL procedure successfully completed.

Compiler - Log Welcome Page ge 7_Modify_Phone_Cursor_with_SYS_REFCURSOR.sql 1_CURSOR_GETPHONES.sql

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vidPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := 'Office';
    phoneCursorSys SYS_REFCURSOR;
    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    getPhonesWithSysRefCursor(vidPerson, vcTypeName, phoneCursorSys);
    LOOP
        FETCH phoneCursorSys INTO vFirstName, vLastName, vnPhoneNumber, vTypeName;
        EXIT WHEN phoneCursorSys%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD(vTypeName, 20));
    END LOOP;
    CLOSE phoneCursorSys;
END;

```

Script Output x

Task completed in 0.101 seconds

Diego Mora	89512342	Office
------------	----------	--------

Compiler - Log x Welcome Page x ge x 7_Modify_Phone_Cursor_with_SYS_REFCURSOR.sql x 1_CURSOR_GETPHONES.sql x

SQL Worksheet History

Worksheet Query Builder

```

--3
DECLARE
    vIdPerson NUMBER(6) := 1;
    vcTypeName VARCHAR2(20) := 'Personal';
    phoneCursorSys SYS_REFCURSOR;
    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    getPhonesWithSysRefCursor(vIdPerson, vcTypeName, phoneCursorSys);
    LOOP
        FETCH phoneCursorSys INTO vFirstName, vLastName, vnPhoneNumber, vTypeName;
        EXIT WHEN phoneCursorSys%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD
    END LOOP;
    CLOSE phoneCursorSys;
END;

```

Script Output x

Task completed in 0.106 seconds

Diego Mora	67576373	Personal
Diego Mora	60518043	Personal

PL/SQL procedure successfully completed.

Compiler - Log Welcome Page ge 7_Modify_Phone_Cursor_with_SYS_REFCURSOR.sql 1_CURSOR_GETPHONES.sql

SQL Worksheet History

Worksheet Query Builder

```
-- 4
DECLARE
    vIdPerson NUMBER(6) := 2;
    vcTypeName VARCHAR2(20) := NULL;
    phoneCursorSys SYS_REFCURSOR;
    vFirstName person.first_name%TYPE;
    vLastName person.first_lastname%TYPE;
    vnPhoneNumber phone.phone_number%TYPE;
    vTypeName phone_type.type_name%TYPE;
BEGIN
    getPhonesWithSysRefCursor(vIdPerson, vcTypeName, phoneCursorSys);
    LOOP
        FETCH phoneCursorSys INTO vFirstName, vLastName, vnPhoneNumber, vTypeName;
        EXIT WHEN phoneCursorSys%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(LPAD(vFirstName || ' ' || vLastName, 20) || LPAD(vnPhoneNumber, 20) || LPAD(vTypeName, 20));
    END LOOP;
    CLOSE phoneCursorSys;
END;
```

Script Output x

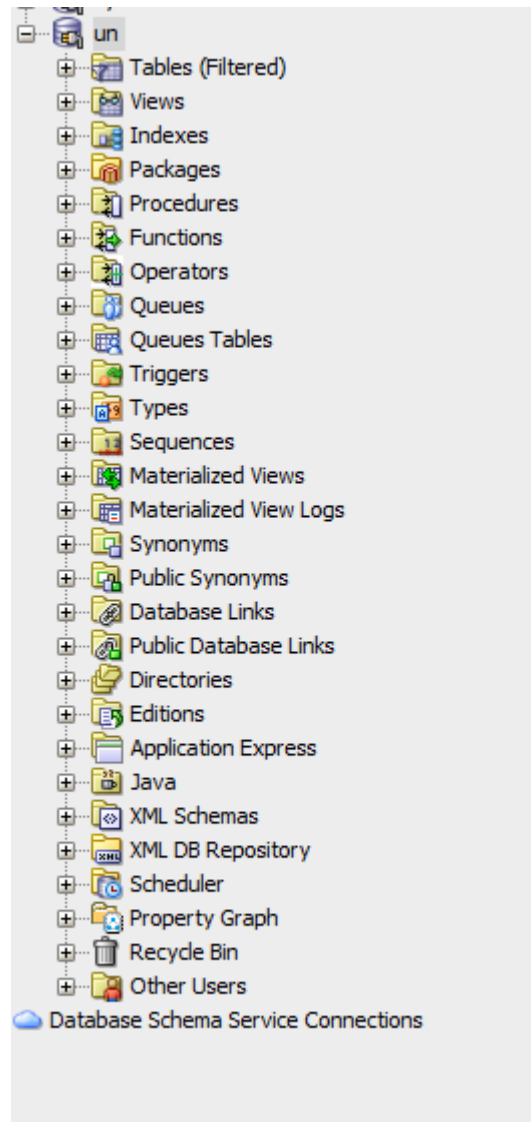
Task completed in 0.137 seconds

Elena Diaz	23515921	Home	
Elena Diaz	80519099	Personal	
Elena Diaz	80519099	Personal	
Elena Diaz	81819039	Office	
Elena Diaz	80001993	Personal	

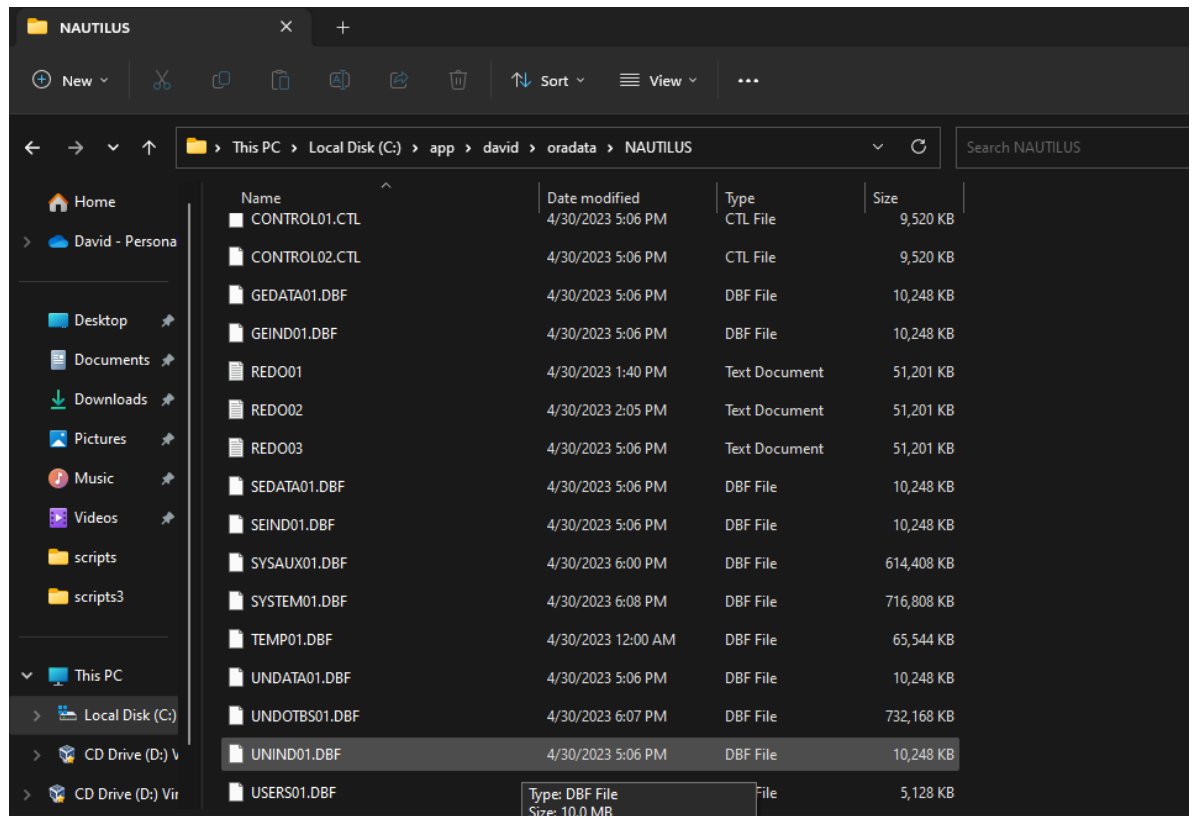
PL/SQL procedure successfully completed.

8. UN Scheme, student table, course table... and getCourses (cursor).

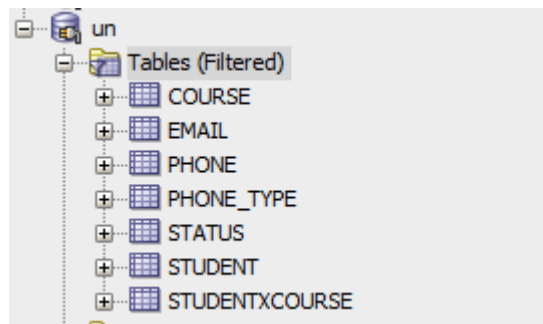
SCHEMA:



TABSPACE:



TABLES:



-STUDENT TABLE

Compiler - LogWelcome Pagegeun8.UN_schema_and_getCourses.sqlSTUDENT

ColumnsDataModelConstraintsGrantsStatisticsTriggersFlashbackDependenciesDetailsPartitionsIndexesSQL

-EMAIL TABLE

Compiler - Log x Welcome Page x SQL ge x SQL un x 8.UN_schema_and_getCourses.sql x EMAIL x

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL

Sort.. Filter:

ID_EMAIL	ID_STUDENT	EMAIL_ADDRESS
1	2	1 josuemora@estudiantec.cr
2	3	2 mariadiaz@estudiantec.cr
3	4	3 josezambranoa@estudiantec.cr

-PHONE TABLE

Compiler - Log x Welcome Page x SQL ge x SQL un x 8.UN_schema_and_getCourses.sql x PHONE x

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL

Sort.. Filter:

ID_PHONE	ID_STUDENT	ID_TYPE	PHONE_NUMBER
1	2	1	1 86324577
2	3	2	1 88324507
3	4	2	2 22324500

- PHONE_TYPE TABLE

Welcome Page x SQL un x 8.UN_schema_and_getCourses.sql x PHONE_TYPE x

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL

Sort.. Filter:

ID_TYPE	TYPE_NAME
1	1 CellPhone
2	2 Home

-STATUS TABLE

Welcome Page x SQL un x 8.UN_schema_and_getCourses.sql x STATUS x

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL

Sort.. Filter:

ID_STATUS	DESCRIPTION
1	1 Approved
2	2 Failed
3	3 Pending

-COURSE TABLE

Welcome Page x un x 8.UN_schema_and_getCourses.sql x STUDENTXCOURSE x				
Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL				
Sort.. Filter:				
	ID_STUDENXCOURSE	ID_STUDENT	ID_COURSE	ID_STATUS
1	1	1	1	1
2	2	1	2	1
3	3	1	3	1
4	4	1	4	1
5	5	1	5	1
6	6	1	6	1
7	7	1	7	1
8	8	1	8	1
9	9	1	9	1
10	10	1	10	1
11	11	1	11	1
12	12	1	12	3
13	13	2	1	1
14	14	2	2	1
15	15	2	3	1
16	16	2	4	1
17	17	2	5	1
18	18	2	6	1
19	19	2	7	1
20	20	2	8	3
21	21	2	9	3
22	22	2	10	3
23	23	2	11	2
24	24	2	12	2
25	25	3	1	1
26	26	3	2	1
27	27	3	3	1
28	28	3	4	1
29	29	3	5	1

CURSOR:

Worksheet	Query Builder
	<pre> CREATE OR REPLACE PROCEDURE getCursos(pIdStudent IN student.id_student%TYPE, pStatus IN status.description%TYPE DEFAULT NULL) AS CURSOR courseCursor IS SELECT s.first_name, s.second_name, s.first_lastname, s.second_lastname, c.course_name, st.description FROM student s INNER JOIN studentxcourse sc ON s.id_student = sc.id_student INNER JOIN course c ON sc.id_course = c.id_course INNER JOIN status st ON sc.id_status = st.id_status WHERE s.id_student = pIdStudent AND (st.description = NVL(pStatus, st.description)); vcFirstName student.first_name%TYPE; vcSecondName student.second_name%TYPE; vcFirstLastname student.first_lastname%TYPE; vcSecondLastname student.second_lastname%TYPE; vcCourseName course.course_name%TYPE; vcIdstatus status.description%TYPE; BEGIN OPEN courseCursor; LOOP FETCH courseCursor INTO vcFirstName, vcSecondName, vcFirstLastname, vcSecondLastname, vcCourseName, vcIdstatus; EXIT WHEN courseCursor%NOTFOUND; DBMS_OUTPUT.PUT_LINE(vcFirstName ' ' vcSecondName ' ' vcFirstLastname ' ' vcSecondLastname ' ' vcCourseName ' ' vcIdstatus); END LOOP; CLOSE courseCursor; END; </pre>

TESTS:

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vIdstudent NUMBER(6) := 1;
    vcstatus VARCHAR2(20) := 'Approved';

BEGIN
    getCursos(vIdstudent,vcstatus);

END;

--2

DECLARE
    vIdStudent student.id_student%TYPE := 1;
    vStatusA status.description%TYPE := 'Approved';

```

Script Output x

Task completed in 0.821 seconds

```

Diego Josue Mora Araya | Introduction to Programming |Approved
Diego Josue Mora Araya | Differential Equations |Approved
Diego Josue Mora Araya | Microcomputer Architecture |Approved
Diego Josue Mora Araya | Microprocessors |Approved
Diego Josue Mora Araya | Math |Approved
Diego Josue Mora Araya | Data Structure |Approved
Diego Josue Mora Araya | English |Approved
Diego Josue Mora Araya | Object-oriented programming |Approved
Diego Josue Mora Araya | Databases I |Approved
Diego Josue Mora Araya | Databases II |Approved
Diego Josue Mora Araya | Software design |Approved

PL/SQL procedure successfully completed.

```


SQL Worksheet | History

Worksheet | Query Builder

```
END;

DECLARE
    vIdstudent NUMBER(6) := 1;
    vcstatus VARCHAR2(20) := 'Pending';

BEGIN
    getCursos(vIdstudent,vcstatus);
END;
```

Script Output x

Task completed in 0.101 seconds

Diego Josue Mora Araya | Project management | Pending

PL/SQL procedure successfully completed.

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vIdstudent NUMBER(6) := 1;

BEGIN
    getCursos(vIdstudent, NULL);
END;
--2

DECLARE
    vIdStudent student.id_student%TYPE := 1;
    vStatusA status.description%TYPE := 'Approved';
    vStatusF status.description%TYPE := 'Failed';
    vStatusP status.description%TYPE := 'Pending';
    vOutput VARCHAR2(4000);
BEGIN
    -- Test when pStatus is not null

```

Script Output x

Task completed in 0.068 seconds

```

Diego Josue Mora Araya | Introduction to Programming |Approved
Diego Josue Mora Araya | Differential Equations |Approved
Diego Josue Mora Araya | Microcomputer Architecture |Approved
Diego Josue Mora Araya | Microprocessors |Approved
Diego Josue Mora Araya | Math |Approved
Diego Josue Mora Araya | Data Structure |Approved
Diego Josue Mora Araya | English |Approved
Diego Josue Mora Araya | Object-oriented programming |Approved
Diego Josue Mora Araya | Databases I |Approved
Diego Josue Mora Araya | Databases II |Approved
Diego Josue Mora Araya | Software design |Approved
Diego Josue Mora Araya | Project management |Pending

PL/SQL procedure successfully completed.

```

SQL Worksheet | History

Worksheet | Query Builder

```
getCursos(vIdstudent,NULL);
END;
--2
DECLARE
    vIdstudent NUMBER(6) := 2;
    vcstatus VARCHAR2(20) := 'Approved';
BEGIN
    getCursos(vIdstudent,vcstatus);
END;
DECLARE
    vIdstudent NUMBER(6) := 2;
    vcstatus VARCHAR2(20) := 'Failed';
```

Script Output x

Task completed in 0.105 seconds

Elena Maria Diaz Mora		Introduction to Programming		Approved
Elena Maria Diaz Mora		Differential Equations		Approved
Elena Maria Diaz Mora		Microcomputer Architecture		Approved
Elena Maria Diaz Mora		Microprocessors		Approved
Elena Maria Diaz Mora		Math		Approved
Elena Maria Diaz Mora		Data Structure		Approved
Elena Maria Diaz Mora		English		Approved

PL/SQL procedure successfully completed.

SQL Worksheet History

Worksheet Query Builder

```
END;  
  
DECLARE  
    vIdstudent NUMBER(6) := 2;  
    vcstatus VARCHAR2(20) := 'Failed';  
  
BEGIN  
    getCursos(vIdstudent,vcstatus);  
  
END;
```

```
DECLARE  
    vIdstudent NUMBER(6) := 2;  
    vcstatus VARCHAR2(20) := 'Pending';
```

Script Output x

Task completed in 0.11 seconds

Elena Maria Diaz Mora | Software design |Failed
Elena Maria Diaz Mora | Project management |Failed

PL/SQL procedure successfully completed.

SQL Worksheet | History

Worksheet | Query Builder

```
END;  
  
DECLARE  
    vIdstudent NUMBER(6) := 2;  
    vcstatus VARCHAR2(20) := 'Pending';  
  
BEGIN  
    getCursos(vIdstudent,vcstatus);  
END;  
  
DECLARE  
    vIdstudent NUMBER(6) := 2;  
  
BEGIN  
    getCursos(vIdstudent,NULL);
```

Script Output x

Task completed in 0.078 seconds

```
Elena Maria Diaz Mora | Object-oriented programming |Pending  
Elena Maria Diaz Mora | Databases I |Pending  
Elena Maria Diaz Mora | Databases II |Pending  
  
PL/SQL procedure successfully completed.
```

SQL Worksheet History

Worksheet Query Builder

```

DECLARE
    vIdstudent NUMBER(6) := 2;
    vcstatus VARCHAR2(20) := 'Pending';

BEGIN
    getCursos(vIdstudent,vcstatus);
END;

DECLARE
    vIdstudent NUMBER(6) := 2;

BEGIN
    getCursos(vIdstudent,NULL);
END;

```

Script Output x

Task completed in 0.105 seconds

Elena Maria Diaz Mora	Introduction to Programming	Approved
Elena Maria Diaz Mora	Differential Equations	Approved
Elena Maria Diaz Mora	Microcomputer Architecture	Approved
Elena Maria Diaz Mora	Microprocessors	Approved
Elena Maria Diaz Mora	Math	Approved
Elena Maria Diaz Mora	Data Structure	Approved
Elena Maria Diaz Mora	English	Approved
Elena Maria Diaz Mora	Object-oriented programming	Pending
Elena Maria Diaz Mora	Databases I	Pending
Elena Maria Diaz Mora	Databases II	Pending
Elena Maria Diaz Mora	Software design	Failed
Elena Maria Diaz Mora	Project management	Failed

PL/SQL procedure successfully completed.

SQL Worksheet History

Worksheet Query Builder

```
--3
DECLARE
    vIdstudent NUMBER(6) := 3;
    vcstatus VARCHAR2(20) := 'Approved';

BEGIN
    getCursos(vIdstudent,vcstatus);

END;
```

```
DECLARE
    vIdstudent NUMBER(6) := 3;
    vcstatus VARCHAR2(20) := 'Failed';

BEGIN
```

Script Output x

Task completed in 0.105 seconds

Antonio Jose Zambrano Flores		Introduction to Programming		Approved
Antonio Jose Zambrano Flores		Differential Equations		Approved
Antonio Jose Zambrano Flores		Microcomputer Architecture		Approved
Antonio Jose Zambrano Flores		Microprocessors		Approved
Antonio Jose Zambrano Flores		Math		Approved

PL/SQL procedure successfully completed.

SQL Worksheet | History | 0.191 seconds

Worksheet | Query Builder

```
BEGIN
    getCursos(vIdstudent,vcstatus);

END;

DECLARE
    vIdstudent NUMBER(6) := 3;
    vcstatus VARCHAR2(20) := 'Failed';

BEGIN
    getCursos(vIdstudent,vcstatus);

END;
```

Script Output x | Task completed in 0.191 seconds

Antonio Jose Zambrano Flores	Data Structure	Failed
Antonio Jose Zambrano Flores	English	Failed
Antonio Jose Zambrano Flores	Object-oriented programming	Failed

PL/SQL procedure successfully completed.

SQL Worksheet History

Worksheet Query Builder

```
vcstatus VARCHAR2(20) := 'Failed';

BEGIN
  getCursos(vIdstudent,vcstatus);

END;
```

```
DECLARE
  vIdstudent NUMBER(6) := 3;
  vcstatus VARCHAR2(20) := 'Pending';

BEGIN
  getCursos(vIdstudent,vcstatus);

END;
```

```
DECLARE
```

Script Output x

Task completed in 0.091 seconds

Antonio Jose Zambrano Flores	Databases I	Pending
Antonio Jose Zambrano Flores	Databases II	Pending
Antonio Jose Zambrano Flores	Software design	Pending
Antonio Jose Zambrano Flores	Project management	Pending

PL/SQL procedure successfully completed.

SQL Worksheet | History

Worksheet | Query Builder

```

DECLARE
    vIdstudent NUMBER(6) := 3;
    vcstatus VARCHAR2(20) := 'Pending';

BEGIN
    getCursos(vIdstudent,vcstatus);
END;

DECLARE
    vIdstudent NUMBER(6) := 3;

BEGIN
    getCursos(vIdstudent,NULL);
END;

```

Script Output x

Task completed in 0.091 seconds

Antonio Jose Zambrano Flores	Introduction to Programming	Approved
Antonio Jose Zambrano Flores	Differential Equations	Approved
Antonio Jose Zambrano Flores	Microcomputer Architecture	Approved
Antonio Jose Zambrano Flores	Microprocessors	Approved
Antonio Jose Zambrano Flores	Math	Approved
Antonio Jose Zambrano Flores	Data Structure	Failed
Antonio Jose Zambrano Flores	English	Failed
Antonio Jose Zambrano Flores	Object-oriented programming	Failed
Antonio Jose Zambrano Flores	Databases I	Pending
Antonio Jose Zambrano Flores	Databases II	Pending
Antonio Jose Zambrano Flores	Software design	Pending
Antonio Jose Zambrano Flores	Project management	Pending

PL/SQL procedure successfully completed.